Kannika Armstrong (Jaikham)

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LinkedIn: kannika-armstrong | GitHub: A-Kannika | Portfolio: KannikaA

SKILLS

Programming Languages: Java, Python, JavaScript, SQL, PostgreSQL, MySQL, HTML, CSS, PHP, R, Rust
Frameworks & Libraries: scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, Streamlit, ReportLab, Plotly
Tools: MS SQL Server, Bootstrap, WordPress, Git, GitHub, Google Colab, Jupyter Notebook, GitHub Actions, Tableau, Excel, VS Code, Linux, Mac OSX, Windows.

EDUCATION

M.S. in Computer Science & Systems | University of Washington, Tacoma, WA | *Expected June 2026*

- Relevant coursework: Advanced Algorithms, Machine Learning, Applied Distributed Computing, Cryptography, Database Systems Internals

B.S. in Computer Science and Systems – University of Washington Tacoma
B.S. in Physics (Minor in Mathematics) – Chiang Mai University, Thailand

PROJECTS, RESEARCH & OPEN SOURCES

Portfolio website: https://a-kannika.github.io/KannikaA/ (for additional information and projects)

Creator/Developer | Vizolytic (Sales Analytics & Reporting Dashboard): GitHub
"A Python and Streamlit-based web application for interactive sales analytics and automated PDF reporting."

- Architected and deployed a web application from scratch using Streamlit and Python to provide interactive sales analytics. (Live at: vizolytic.streamlit.app)
- Built interactive Plotly visualizations, including U.S. choropleth maps and hierarchical treemaps.
- Engineered data filtering, time series analysis, and CSV export features using Pandas.
- Developed an automated PDF reporting feature (ReportLab) to summarize key business insights.

Empirical Analysis of Network Flow Algorithms: GitHub
"A Java-based empirical study comparing the performance of classic and optimized network flow algorithms."

- Directed a 4-developer team to implement and analyze the performance of complex network flow algorithms.
- Implemented multiple max-flow algorithms in Java, including Ford-Fulkerson, Scaling Ford-Fulkerson, and Preflow-Push.
- Engineered a system to parse graph data from files and dynamically manage residual graphs during execution.
- Benchmarked algorithm runtimes against diverse graph structures to identify and document performance trade-offs.

Master's Thesis | Optimizing Performance and Efficiency of Lattice-Based Post-Quantum Cryptography Algorithms in Resource-Constrained UAV Environments (In Progress) *University of Washington Tacoma*
"Developing, optimizing, and verifying a secure, quantum-resistant protocol for UAVs."

- Architecting a secure UAV protocol by integrating Kyber (PQC) and Ascon-MAC.
- Implementing NTT, vectorization, and parallelization to optimize for resource-constrained hardware.
- Simulating protocol performance in Python to benchmark latency, memory, and computation time.
- Conducting formal security verification with ProVerif to ensure quantum resistance and protocol robustness.

PROFESSIONAL EXPERIENCE

Teaching Assistant & Academic Mentor | University of Washington & Pierce College *Sep 2024 – Jun 2025 (UW); 2020 – 2024 (Pierce)*

- Led lab sessions, graded Java programming projects, and mentored students in debugging.
- Mentored 50+ students in Algorithms, OOP (Java/Python), and SQL queries.
- Taught collaborative project workflows using Git, GitHub, and VS Code.

Meteorologist & Secretary to the Deputy Director-General for Academic Affairs | Thai Meteorological Dept., Thailand
May 2006 – May 2018

- Developed predictive models and analyzed large-scale climate data to provide actionable insights.
- Coordinated inter-agency academic projects, advised executive leadership, and managed project lifecycles by reviewing, summarizing, and drafting technical reports and policy.